THE LAG-TWO SHARK-FIN OBSERVABLE FOR RIEMANN-ZETA ZEROS: EXACT CLOCK CANCELLATION, COUNTER-BAND REDUCTION, SUPERFINE PHASE SENSITIVITY, AND A TWO-MILLION-ZERO AUDIT

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ABSTRACT. Let γ_k be the ordinates of the nontrivial zeros of the Riemann zeta function and let τ_k be the smooth Riemann–von Mangoldt clock defined by $M(\tau_k) = k - 1/2$. With the rapidly growing index frequency $q_k = k \log k$, we study $a_k = \exp(\mathrm{i} q_k \gamma_k) - \exp(\mathrm{i} q_k \tau_k)$ and the fixed lag-two increment $\Delta_{k,2} = a_{k+2} - a_k$. If the four constituent phases were independent Haar phases, the radius $|\Delta_{k,2}|$ would follow the radial law of a four-step planar uniform random walk, called the shark-fin law here.

We separate the rigorous deterministic part of this phenomenon from the remaining arithmetic phase problem. First, the clock is inverted exactly by Lambert W, and the lag-two clock correlation is shown to vanish by a Kusmin–Landau estimate after an exact cancellation of the quadratic phase. Second, a block-transfer lemma reduces the second radial moment to two mesoscopic phase-cancellation statements. Third, an oriented-integral formula replaces contour displacement by a short band between the true and smooth counting functions. Fourth, the residual phase increment is expressed exactly through two consecutive normalized gaps and the accumulated counting remainder. Finally, a scale-invisibility theorem proves that ordinary weak gap laws do not determine this observable: perturbations of size $1/(k \log k)$ preserve fixed-dimensional gap statistics at their natural scale while prescribing the shark-fin phases arbitrarily.

The complete first 2,001,052 zero ordinates are used for a numerical audit. The first five even radial moments agree with the exact four-step moments $4, 28, 256, 2716, 31504$; the relevant phase block means follow the $L^{-1/2}$ law from $L = 4$ through $L = 4096$; and the lag-two residual phase is angularly uniform at the resolution of the reported tests. The paper thereby identifies both the proved component and the precise superfine theorem required to complete the asymptotic shark-fin law for the actual zeta zeros.

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1. Introduction

The local statistics of the Riemann-zeta zeros are commonly studied after normalization by the mean spacing. Montgomery's pair-correlation theorem and conjecture initiated the comparison with random-matrix spectra [14]; the higher-correlation framework of Rudnick and Sarnak [17] and the numerical work of Odlyzko [15, 16] developed this viewpoint further. The observable considered here is substantially more sensitive. It multiplies the k-th ordinate by $q_k = k \log k$, so that a displacement of order $1/(k \log k)$ already changes its phase by order one.

The purpose of this paper is to give a complete account of the fixed-lag shark-fin calculation developed for $h = 1, 2$, with the rigorous analysis concentrated on $h = 2$. The main conclusions are as follows.

(i) The four-step radial law and all of its even moments are explicit.
(ii) The deterministic clock-clock correlation vanishes for every fixed lag.
(iii) For $h = 2$, the second radial moment is reduced to two concrete mesoscopic phase averages.
(iv) The same reduction can be written without a contour near the critical line, using only the band between two counting functions.
(v) The internal clock resonance is 4π. The Lambert correction breaks it at size $(\log \log k)/ \log k$.
(vi) The remaining phase is exactly a function of two consecutive normalized gaps and the Riemann–von Mangoldt remainder.
(vii) Ordinary gap laws are too coarse to force the required cancellation.

Items (ii)–(vi) are exact reductions or theorems. Item (vii) is established by an explicit perturbation theorem. The asymptotic cancellation for the true zero phases at the final superfine scale is recorded as a conjecture, while the finite-height evidence is stated only in experiment environments.

2. The smooth clock and the lagged observable

Definition 2.1 (Riemann–von Mangoldt clock)**.** Set

$$M(t) = \frac{t}{2\pi} \log \frac{t}{2\pi \mathrm{e}} + \frac{7}{8}. \tag{2.1}$$

For each positive integer k, let τ_k denote the solution of

$$M(\tau_k) = k - \frac{1}{2}. \tag{2.2}$$

The function M is the smooth main term in the Riemann–von Mangoldt formula; see [18, Ch. 9].

Definition 2.2 (Phase displacement)**.** Let

$$q_k = k \log k, \tag{2.3}$$

$$F_k = q_k \tau_k, \tag{2.4}$$

$$E_k = q_k(\gamma_k - \tau_k), \tag{2.5}$$

$$w_k = \exp(\mathrm{i}F_k), \qquad b_k = \exp(\mathrm{i}E_k), \tag{2.6}$$

$$z_k = \exp(\mathrm{i}q_k\gamma_k) = w_k b_k, \tag{2.7}$$

$$a_k = z_k - w_k = w_k(b_k - 1). \tag{2.8}$$

For a fixed positive lag h, define

$$\Delta_{k,h} = a_{k+h} - a_k. \tag{2.9}$$

The expansion

$$\Delta_{k,h} = \mathrm{e}^{\mathrm{i}q_{k+h}\gamma_{k+h}} - \mathrm{e}^{\mathrm{i}q_{k+h}\tau_{k+h}} - \mathrm{e}^{\mathrm{i}q_k\gamma_k} + \mathrm{e}^{\mathrm{i}q_k\tau_k} \tag{2.10}$$

contains four unit complex numbers with fixed signs. This is the source of the four-step random-walk law.

3. The four-step radial law

Definition 3.1 (Four-step shark-fin law)**.** Let U_1, U_2, U_3, U_4 be independent Haar random variables on the unit circle. The shark-fin radius is

$$R_4 = |U_1 + U_2 + U_3 + U_4|. \tag{3.1}$$

Its density on $[0, 4]$ is denoted by $p_4(r)$. In the Bessel-transform normalization used for planar uniform random walks,

$$p_4(r) = r \int_0^\infty J_0(rt) J_0(t)^4 t \, dt, \qquad 0 < r < 4, \tag{3.2}$$

with the integral interpreted in the standard oscillatory sense. Short uniform random-walk densities and their singularities are treated in [2, 3].

Proposition 3.2 (Even moments)**.** *For every integer $m \geq 0$,*

$$C_m := \mathbb{E}R_4^{2m} = \sum_{a_1+a_2+a_3+a_4=m} \binom{m}{a_1, a_2, a_3, a_4}^2. \tag{3.3}$$

In particular,

$$(C_1, C_2, C_3, C_4, C_5) = (4, 28, 256, 2716, 31504). \tag{3.4}$$

Proof. Write $S = U_1 + \cdots + U_4$. Expanding $S^m \overline{S}^{,m}$, Haar orthogonality forces the exponent of each U_j in the first factor to equal its exponent in the conjugate factor. The surviving coefficient attached to (a_1, a_2, a_3, a_4) is the square of the multinomial coefficient, yielding (3.3). The values in (3.4) follow by direct summation. □

Corollary 3.3 (Moment determination)**.** *If, for a fixed lag h,*

$$\frac{1}{N} \sum_{k \leq N} |\Delta_{k,h}|^{2m} \longrightarrow C_m \qquad (m \geq 1), \tag{3.5}$$

then the empirical radial measures of $|\Delta_{k,h}|$ converge weakly to $p_4(r)\, dr$.

Proof. All measures are supported on the compact interval $[0, 4]$. Polynomials in r^2 are dense in the continuous functions of r on this interval after the substitution $x = r^2$. Thus the moment limits determine the weak limit. □

3.1. **The factor $1-\operatorname{sinc}^2(\pi u)$.** We use $\operatorname{sinc} x = \sin(x)/x$, with $\operatorname{sinc} 0 = 1$. Expressions of the form

$$p_4(r)\big[1-\operatorname{sinc}^2(\pi u)\big] \tag{3.6}$$

occur when a radial four-step law is multiplied by a lag- or frequency-window factor. The factor in brackets is not part of the definition of p_4. Indeed, integrating (3.6) in r gives

$$\int_0^4 p_4(r)\big[1-\operatorname{sinc}^2(\pi u)\big]\,dr = 1-\operatorname{sinc}^2(\pi u), \tag{3.7}$$

so (3.6) is a weighted surface rather than a normalized conditional density in r. In particular it vanishes at $u=0$. A derivation of such a window factor does not establish the four-phase equidistribution needed for (3.5).

4. The full Weyl criterion and why fixed-frequency theorems stop short

If the four phases in (2.10) were jointly Haar equidistributed, the radial law would be p_4. Weyl's criterion [19] makes the exact requirement explicit.

Proposition 4.1 (Four-phase Weyl reduction)**.** *For $m=(m_1,m_2,m_3,m_4)\in\mathbb{Z}^4$, define*

$$\Phi_{m,h}(k) = (m_1+m_3)F_{k+h} + (m_2+m_4)F_k + m_1E_{k+h} + m_2E_k. \tag{4.1}$$

Joint equidistribution of the four phases follows if and only if

$$\frac{1}{N}\sum_{k\le N}\exp\big(\mathrm{i}\Phi_{m,h}(k)\big)\longrightarrow 0 \tag{4.2}$$

for every nonzero $m\in\mathbb{Z}^4$.

Two necessary subfamilies are

$$\frac{1}{N}\sum_{k\le N}\mathrm{e}^{\mathrm{i}E_{k+h}}\longrightarrow 0, \tag{4.3}$$

$$\frac{1}{N}\sum_{k\le N}\mathrm{e}^{\mathrm{i}(E_{k+h}-E_k)}\longrightarrow 0. \tag{4.4}$$

Hlawka proved uniform distribution results for fixed multiples of zero ordinates [11]; Fujii developed broad fixed-function criteria [9]; and later work refined simultaneous distribution questions [8, 7, 5]. These theorems do not directly cover the triangular phase $q_k\gamma_k$ because the multiplier itself grows as $k\log k$.

Replacing k by $M(\gamma_k)$ does not remove the issue. The smooth substitute has size

$$\gamma\,M(\gamma)\log M(\gamma) \asymp \frac{\gamma^2}{2\pi}(\log\gamma)^2, \tag{4.5}$$

while the discrepancy between k and $M(\gamma_k)$ is precisely the fluctuation whose amplified phase is under study.

The central-limit theorems of Selberg type and their zero-displacement versions also use fixed test functions or controlled mesoscopic regimes. Kargin proves Gaussian fluctuations and a two-zero theorem when the index separation grows as $(\log k)^\beta$, $\beta>0$, rather than remaining equal to one or two [12]. Bourgade's mesoscopic fluctuation theorem similarly concerns statistics at a coarser scale [4]. Weak convergence of a normalized variable X_k cannot by itself control $\mathbb{E}\exp(\mathrm{i}t_kX_k)$ when $t_k\to\infty$.

4.1. **Why the paper concentrates on $h = 2$.** For both $h = 1$ and $h = 2$, the Weyl requirements (4.3)–(4.4) involve fixed-neighbor residual phases at growing frequency. The lag-two case has two additional structures developed in this paper: its remainder increment is the sum of exactly two normalized gaps, and the clock derivative has the principal resonance 4π. It is also the lag for which the BLS variance relation was proposed as a possible bridge. The fixed-lag clock theorem below covers $h = 1$ as well, but the counter-band kernel, two-gap formula, and numerical tables are specialized to $h = 2$.

5. Exact scalar reduction of the residual phase

Definition 5.1 (Counting remainder)**.** Set

$$R_k = k - \frac{1}{2} - M(\gamma_k), \tag{5.1}$$

$$d_k = \gamma_k - \tau_k, \tag{5.2}$$

$$L_k = \log \frac{\tau_k}{2\pi}, \tag{5.3}$$

$$\overline{L}_k = \int_0^1 \log \frac{\tau_k + s d_k}{2\pi}\, ds. \tag{5.4}$$

Proposition 5.2 (Exact residual formula)**.** *For every k,*

$$d_k = -\frac{2\pi R_k}{\overline{L}_k}, \tag{5.5}$$

$$E_k = -2\pi k \frac{\log k}{\overline{L}_k} R_k. \tag{5.6}$$

Proof. Since

$$M'(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}, \tag{5.7}$$

the fundamental theorem of calculus gives

$$M(\gamma_k) - M(\tau_k) = \frac{d_k}{2\pi} \overline{L}_k. \tag{5.8}$$

The left-hand side equals $-R_k$ by (2.2) and (5.1). This proves (5.5); multiplication by q_k proves (5.6). □

The clock equation also gives

$$k = \frac{\tau_k}{2\pi}(L_k - 1) + \frac{11}{8}. \tag{5.9}$$

Consequently, with

$$A_k = 2\pi k \left(1 + \frac{\log(L_k - 1)}{L_k}\right), \tag{5.10}$$

one obtains the asymptotic reduction

$$E_k = -A_k R_k + O\left(\frac{R_k^2 + |R_k|}{L_k}\right). \tag{5.11}$$

On dyadic windows where the fixed second moments of R_k have their known Selberg–Kargin size, the mean of the error in (5.11) tends to zero. The inequality $|e^{ix} - e^{iy}| \le |x - y|$ then transfers the single-phase problem to

$$\frac{1}{N} \sum_{k \le N} e^{-iA_k R_k}. \tag{5.12}$$

The reduction is useful because it is scalar and exact up to a mean-small term; it does not lower the effective frequency $A_k \asymp k$.

Warning 5.3 (Prime-polynomial substitution). If P_k approximates R_k, the elementary estimate

$$|\mathrm{e}^{-\mathrm{i}A_kR_k} - \mathrm{e}^{-\mathrm{i}A_kP_k}| \le A_k|R_k - P_k| \tag{5.13}$$

would require, on $k \asymp N$,

$$\frac{1}{N}\sum_{N<k\le 2N} |R_k - P_k| = o(N^{-1}). \tag{5.14}$$

Fixed-moment Selberg approximations do not supply this precision. Thus a short prime polynomial cannot simply be inserted into (5.12) by a Lipschitz argument.

6. The BLS variance route at lag two

Define the two-spacing length in smooth coordinates by

$$S_{k,2} = M(\gamma_{k+2}) - M(\gamma_k). \tag{6.1}$$

The exact telescoping identity is

$$R_{k+2} - R_k = 2 - S_{k,2}. \tag{6.2}$$

Bohigas, Leboeuf, and Sánchez relate spacing variances and number variances in random-matrix and spectral models [1]. In the notation relevant here, their large-index relation has the form

$$\sigma^2(n) \approx \Sigma^2(n) - \frac{1}{6}, \tag{6.3}$$

and the number variance is represented through a form factor by an integral of the type

$$\Sigma^2(L) = \frac{2}{\pi^2}\int_0^\infty K(\tau)\frac{\sin^2(\pi L\tau)}{\tau^2}\,d\tau. \tag{6.4}$$

Proposition 6.1 (What the variance does not determine). *Knowledge of $\Sigma^2(2)$, or equivalently of a second moment inferred from* (6.3)*, does not determine the characteristic function*

$$\mathbb{E}\mathrm{e}^{-\mathrm{i}tS_{k,2}} \tag{6.5}$$

at either fixed or growing t.

Proof. Mean and variance impose only two linear constraints on a probability measure. Distinct compactly supported measures may have the same first two moments and different Fourier transforms at any prescribed nonzero frequency. The phase problem here requires $t \asymp A_k \asymp k$, which is strictly stronger. □

Thus the BLS formula supplies meaningful information about the two-spacing variance but not the growing-frequency phase needed for the shark-fin observable. This distinction prevents a variance identity from being used as a hidden characteristic-function theorem.

7. Direct radial reduction for the second moment

Proposition 7.1 (Exact second-moment decomposition). *For $h = 2$,*

$$\frac{1}{N}\sum_{k\le N}|\Delta_{k,2}|^2 = 4 - 4\,\mathrm{Re}\left(\frac{1}{N}\sum_{k\le N} b_k\right) - 2\,\mathrm{Re}\left(\frac{1}{N}\sum_{k\le N} a_{k+2}\overline{a_k}\right) + o(1). \tag{7.1}$$

Moreover,

$$a_{k+2}\overline{a_k} = z_{k+2}\overline{z_k} - z_{k+2}\overline{w_k} - w_{k+2}\overline{z_k} + w_{k+2}\overline{w_k}. \tag{7.2}$$

Proof. Expand $|a_{k+2} - a_k|^2$, shift the endpoint sums by two indices, and use

$$|a_k|^2 = |b_k - 1|^2 = 2 - 2\operatorname{Re} b_k. \tag{7.3}$$

The second identity follows by expanding $(z_{k+2} - w_{k+2})(\overline{z}_k - \overline{w}_k)$. □

8. Exact Lambert inversion and deterministic clock cancellation

Proposition 8.1 (Lambert inversion)**.** *The continuous clock inverse is*

$$\tau(x) = \frac{2\pi(x - 11/8)}{W((x - 11/8)/\mathrm{e})}. \tag{8.1}$$

Proof. Set $y = \tau/(2\pi)$. Equation (2.2) becomes

$$y(\log y - 1) = x - \frac{11}{8}. \tag{8.2}$$

Writing $v = \log y - 1$ gives $v\mathrm{e}^v = (x - 11/8)/\mathrm{e}$, and inversion by W proves (8.1). We use the standard properties and asymptotics of Lambert W from [6]. □

Define

$$F(x) = x \log x\, \tau(x), \tag{8.3}$$

$$G(x) = F(x) - 2\pi x^2. \tag{8.4}$$

Lemma 8.2 (Lambert remainder)**.** *With $L = \log x$ and $\ell = \log\log x$,*

$$G(x) = 2\pi x^2 \left(\frac{\ell + 1}{L} + O\left(\frac{\ell^2}{L^2} + \frac{1}{x} \right) \right), \tag{8.5}$$

$$G''(x) = 4\pi \frac{\ell + 1}{L} \left(1 + O\left(\frac{\ell}{L} \right) \right). \tag{8.6}$$

Proof. Apply the classical expansion

$$W(y) = \log y - \log\log y + O\left(\frac{\log\log y}{\log y} \right) \tag{8.7}$$

to $y = (x - 11/8)/\mathrm{e}$ in (8.1), and expand the reciprocal. This gives (8.5). For the differentiated formula, differentiate the exact expression (8.1) using

$$W'(y) = \frac{W(y)}{y(1 + W(y))} \tag{8.8}$$

twice before inserting the same expansion. The twice-differentiated leading term $2\pi x^2(\ell + 1)/L$ is $4\pi(\ell+1)/L + O(\ell/L^2)$, and the differentiated remainder is $O(\ell^2/L^2)$, proving (8.6). These operations are also obtained from the complete differentiable asymptotic expansion in [6]. □

Theorem 8.3 (Fixed-lag clock cancellation)**.** *For every fixed positive integer h,*

$$\frac{1}{N} \sum_{k \le N} w_{k+h}\overline{w_k} \longrightarrow 0. \tag{8.9}$$

More precisely,

$$\sum_{k \le N} w_{k+h}\overline{w_k} \ll_h \frac{\log N}{\log\log N} \tag{8.10}$$

after omission of a fixed initial segment.

Proof. The phase is $F(k+h)-F(k)$. The quadratic contribution satisfies

$$2\pi\big((k+h)^2-k^2\big)\in 2\pi\mathbb{Z}, \tag{8.11}$$

so only

$$H_h(x)=G(x+h)-G(x) \tag{8.12}$$

remains. From (8.6),

$$\frac{H_h'(x)}{2\pi}=2h\frac{\log\log x+1}{\log x}(1+o(1)). \tag{8.13}$$

This quantity is eventually positive, decreasing, and below $1/2$. The Kusmin–Landau first-derivative estimate [10, Ch. 2] therefore gives (8.10), and division by N gives (8.9). □

Theorem 8.3 disposes rigorously of the final term in (7.2). Each of the other terms contains at least one true zero ordinate.

9. Mesoscopic block transfer

Set

$$h_k=w_{k+2}\overline{w_k}=\mathrm{e}^{\mathrm{i}H_2(k)}. \tag{9.1}$$

On $N<k\le 2N$, equation (8.13) implies

$$|h_{k+1}-h_k|\ll a_N,\qquad a_N=\frac{\log\log N}{\log N}. \tag{9.2}$$

Lemma 9.1 (Block transfer)**.** *Let $L_N\to\infty$ and $L_Na_N\to 0$. Partition $(N,2N]$ into consecutive blocks I of length L_N. For any unit-modulus sequence (u_k),*

$$\left|\frac{1}{N}\sum_{N<k\le 2N}h_ku_k\right|\le \mathrm{Avg}_I\left|\frac{1}{L_N}\sum_{k\in I}u_k\right| + O(L_Na_N)+O(L_N/N). \tag{9.3}$$

Proof. For each block I, freeze h_k at its left endpoint k_I. The variation estimate (9.2) yields

$$\frac{1}{L_N}\sum_{k\in I}h_ku_k=h_{k_I}\frac{1}{L_N}\sum_{k\in I}u_k+O(L_Na_N). \tag{9.4}$$

Average over the complete blocks and bound the incomplete endpoint block by $O(L_N/N)$. □

Since

$$z_{k+2}\overline{w_k}=h_kb_{k+2}, \tag{9.5}$$

$$w_{k+2}\overline{z_k}=h_k\overline{b_k}, \tag{9.6}$$

$$z_{k+2}\overline{z_k}=h_kc_k^{(2)},\qquad c_k^{(2)}=b_{k+2}\overline{b_k}, \tag{9.7}$$

the lemma gives the following sufficient theorem.

Theorem 9.2 (Mesoscopic criterion for the second radial moment)**.** *Suppose that for some $L_N\to\infty$ satisfying*

$$L_N\frac{\log\log N}{\log N}\longrightarrow 0, \tag{9.8}$$

one has

$$\mathrm{Avg}_I \left| \frac{1}{L_N} \sum_{k \in I} b_k \right| \longrightarrow 0, \tag{9.9}$$

$$\mathrm{Avg}_I \left| \frac{1}{L_N} \sum_{k \in I} c_k^{(2)} \right| \longrightarrow 0. \tag{9.10}$$

Then

$$\frac{1}{N} \sum_{k \le N} |\Delta_{k,2}|^2 \longrightarrow 4. \tag{9.11}$$

Proof. Condition (9.9), Lemma 9.1, and (9.5)–(9.6) eliminate the two mixed correlations. Condition (9.10) and (9.7) eliminate the zero-zero correlation. Theorem 8.3 eliminates the clock-clock correlation. Condition (9.9) also implies the required global mean of b_k is zero. Proposition 7.1 then gives (9.11). □

A convenient admissible scale is

$$L_N = \left\lfloor \sqrt{\frac{\log N}{\log \log N}} \right\rfloor. \tag{9.12}$$

10. A contour-free counter-band formula

The next identity avoids an explicit formula evaluated on or near the critical line. It uses only the short oriented interval between τ_k and γ_k.

Proposition 10.1 (Oriented interval representation)**.** *For every k,*

$$a_k = \mathrm{i} q_k \int_{\tau_k}^{\gamma_k} \mathrm{e}^{\mathrm{i} q_k t}\, dt, \tag{10.1}$$

where the integral is oriented.

Proof. This is the fundamental theorem of calculus applied to $t \mapsto \mathrm{e}^{\mathrm{i} q_k t}$. □

Define the smooth and true counters

$$A(t) = \#\{k : \tau_k \le t\} = \left\lfloor M(t) + \frac{1}{2} \right\rfloor, \tag{10.2}$$

$$N(t) = \#\{k : \gamma_k \le t\}, \tag{10.3}$$

and the oriented indicators

$$\chi_k(t) = \begin{cases} 1, & \tau_k < t \le \gamma_k, \\ -1, & \gamma_k < t \le \tau_k, \\ 0, & \text{otherwise.} \end{cases} \tag{10.4}$$

Then

$$\sum_k \chi_k(t) = A(t) - N(t). \tag{10.5}$$

The active indices form the consecutive interval between $A(t)$ and $N(t)$. Riemann–von Mangoldt gives

$$|A(t) - N(t)| = O(\log t). \tag{10.6}$$

Thus

$$\sum_{k\le K} a_k = \mathrm{i}\int \sum_{k\in I_K(t)} q_k \mathrm{e}^{\mathrm{i}q_k t}\chi_k(t)\,dt, \tag{10.7}$$

where $I_K(t)$ is a consecutive interval of at most $O(\log t)$ indices.

Proposition 10.2 (Lag-two counter-band kernel)**.** *One has*

$$\sum_{k\le K-2} a_{k+2}\overline{a_k} = \iint \mathcal{K}_{2,K}(t,u)\,dt\,du, \tag{10.8}$$

where

$$\mathcal{K}_{2,K}(t,u) = \sum_{k\le K-2} q_{k+2}q_k \mathrm{e}^{\mathrm{i}(q_{k+2}t-q_k u)} \times \chi_{k+2}(t)\chi_k(u). \tag{10.9}$$

At every (t,u), the summation set is

$$(I_K(t)-2)\cap I_K(u) \tag{10.10}$$

and has cardinality

$$O\big(\min(\log t,\log u)\big). \tag{10.11}$$

The remaining analytic target in this representation is

$$\iint \mathcal{K}_{2,K}(t,u)\,dt\,du = o(K), \tag{10.12}$$

together with the diagonal mean of b_k. Absolute estimation of (10.9) loses the oscillation; the index sum must be exploited before the integrations.

11. The internal 4π resonance at lag two

For fixed (t,u), continue the phase in (10.9) to

$$F_{t,u}(x) = q(x+2)t - q(x)u, \qquad q(x) = x\log x. \tag{11.1}$$

Its derivative is

$$F'_{t,u}(x) = t(\log(x+2)+1) - u(\log x+1) \tag{11.2}$$

$$= (t-u)(\log x+1) + \frac{2t}{x} + O\left(\frac{t}{x^2}\right). \tag{11.3}$$

In the central clock region $u\sim\tau_x$ and $t\sim\tau_{x+2}$, while

$$\tau_{x+2}-\tau_x = \frac{4\pi}{\log(\tau_x/(2\pi))} + O\left(\frac{1}{x(\log x)^2}\right). \tag{11.4}$$

The leading value of (11.3) is consequently 4π, an integer multiple of 2π.

The detuning is seen exactly on the clock. Define

$$\Phi_2(x) = q(x+2)\tau_{x+2} - q(x)\tau_x. \tag{11.5}$$

Since $q(x)\tau_x = 2\pi x^2 + G(x)$,

$$\Phi_2(x) = 8\pi x + 8\pi + G(x+2) - G(x). \tag{11.6}$$

The first two terms are invisible after exponentiation, and

$$\Phi_2'(x) = 8\pi\frac{\log\log x+1}{\log x}\left(1+O\left(\frac{\log\log x}{\log x}\right)\right). \tag{11.7}$$

Kusmin–Landau therefore yields, for a block of length L,

$$\left|\sum_{x<k\le x+L} \mathrm{e}^{\mathrm{i}\Phi_2(k)}\right| \ll \min\left(L, \frac{\log x}{\log\log x}\right). \tag{11.8}$$

At the natural band length $L \asymp \log x$, the Lambert detuning saves a factor $\log\log x$.

For true endpoints, the four phases in (7.2) are

$$\Phi_2(k), \tag{11.9}$$

$$\Phi_2(k) + q_{k+2}d_{k+2}, \tag{11.10}$$

$$\Phi_2(k) - q_k d_k, \tag{11.11}$$

$$\Phi_2(k) + q_{k+2}d_{k+2} - q_k d_k. \tag{11.12}$$

Equation (11.8) controls the first. The remaining three require information about the weighted displacement increments.

12. Exact reduction to two consecutive normalized gaps

Definition 12.1 (Normalized gap)**.** Define

$$g_k = M(\gamma_{k+1}) - M(\gamma_k). \tag{12.1}$$

Lemma 12.2 (Two-gap recurrence)**.** *The counting remainder satisfies the exact identity*

$$R_{k+2} - R_k = 2 - g_k - g_{k+1}. \tag{12.2}$$

Proof. Subtract the definitions of R_{k+2} and R_k, and telescope the increment of M. □

Set

$$C_k = 2\pi k \frac{\log k}{\overline{L}_k}, \tag{12.3}$$

so that $E_k = -C_k R_k$ exactly.

Theorem 12.3 (Exact lag-two residual phase)**.** *One has*

$$E_{k+2} - E_k = -C_{k+2}(2 - g_k - g_{k+1}) - (C_{k+2} - C_k)R_k. \tag{12.4}$$

Equivalently,

$$c_k^{(2)} = \exp\left(-\mathrm{i}C_{k+2}(2 - g_k - g_{k+1})\right) \times \exp\left(-\mathrm{i}(C_{k+2} - C_k)R_k\right). \tag{12.5}$$

Proof. Substitute (12.2) into $-C_{k+2}R_{k+2} + C_k R_k$ and collect the coefficient of R_k. □

Theorem 12.3 identifies the minimal local arithmetic object for $h = 2$: two consecutive smooth-normalized gaps together with the accumulated remainder R_k. A variance of $g_k + g_{k+1}$ alone does not determine (12.5).

13. A scale-invisibility theorem

The phase in (2.5) resolves perturbations of order q_k^{-1}. The next theorem shows that ordinary fixed-dimensional gap laws cannot, by themselves, determine such a phase.

Theorem 13.1 (Superfine phase prescription)**.** *Let (x_k) be strictly increasing and suppose that, eventually,*

$$x_{k+1} - x_k > \frac{3\pi}{q_k}. \tag{13.1}$$

For every real sequence (ϑ_k), there exists a strictly increasing sequence $(\widehat{x}_k)$ such that

$$|\widehat{x}_k - x_k| \le \frac{\pi}{q_k}, \tag{13.2}$$

$$q_k(\widehat{x}_k - \tau_k) \equiv \vartheta_k \pmod{2\pi}. \tag{13.3}$$

Furthermore, for each fixed positive integer r,

$$\left|(\widehat{x}_{k+r} - \widehat{x}_k) - (x_{k+r} - x_k)\right| = O\left(\frac{1}{k\log k}\right). \tag{13.4}$$

Thus every bounded uniformly continuous statistic of finitely many gaps, normalized at scale $1/\log k$, has the same asymptotic value for the two sequences whenever either value exists.

Proof. Choose $m_k \in \mathbb{Z}$ so that

$$\left|x_k - \tau_k - \frac{\vartheta_k + 2\pi m_k}{q_k}\right| \le \frac{\pi}{q_k}, \tag{13.5}$$

and define

$$\widehat{x}_k = \tau_k + \frac{\vartheta_k + 2\pi m_k}{q_k}. \tag{13.6}$$

Equations (13.2) and (13.3) follow immediately. Moreover,

$$\widehat{x}_{k+1} - \widehat{x}_k \ge x_{k+1} - x_k - \frac{\pi}{q_{k+1}} - \frac{\pi}{q_k} > 0 \tag{13.7}$$

eventually, so the modified sequence is increasing. For fixed r, the triangle inequality gives

$$\left|(\widehat{x}_{k+r} - \widehat{x}_k) - (x_{k+r} - x_k)\right| \le \frac{\pi}{q_{k+r}} + \frac{\pi}{q_k}, \tag{13.8}$$

which proves (13.4). After multiplication by the local scale $\asymp \log k$, the perturbation is $O(k^{-1})$. Uniform continuity then proves the final statement. □

Corollary 13.2 (Maximal failure at lag two)**.** *Under the hypotheses of Theorem 13.1, choose*

$$\vartheta_{k+2} = \vartheta_k. \tag{13.9}$$

Then the fixed-dimensional normalized gap statistics are unchanged, whereas

$$\mathrm{e}^{\mathrm{i}(\vartheta_{k+2} - \vartheta_k)} = 1 \tag{13.10}$$

for every k.

The theorem separates two resolutions:

$$\text{natural gap scale} \asymp \frac{1}{\log k}, \tag{13.11}$$

$$\text{shark-fin phase scale} = \frac{1}{k\log k}. \tag{13.12}$$

The factor between them is k. This does not assert that the actual zeta zeros admit the adversarial modification while remaining zeros of a zeta function. It proves the logically sharper point that weak gap information at the natural scale does not encode the superfine phase.

Conjecture 13.3 (Lag-two superfine cancellation)**.** *For the actual zeta zeros, uniformly for the mesoscopic blocks required in Theorem 9.2,*

$$\frac{1}{|I|}\sum_{k\in I}\exp\left(-\mathrm{i}C_{k+2}(2-g_k-g_{k+1})\right) \times \exp\left(-\mathrm{i}(C_{k+2}-C_k)R_k\right)=o(1). \tag{13.13}$$

The available fixed-frequency, pair-correlation, and mesoscopic Gaussian theorems cited above do not provide the growing-frequency estimate (13.13). The conjecture is therefore the new arithmetic input isolated by the exact reduction, rather than a renamed form of a variance statement.

14. Numerical audit on 2,001,052 zeros

Experiment 14.1 (Data and computation)**.** The audit uses all 2,001,052 ordinates in the supplied binary table, originating from A. M. Odlyzko's public zero tables [16]. The corresponding τ_k values are deterministic Lambert-clock quantities. All phase calculations use IEEE double precision. No subset is used in the reported aggregate values.

14.1. **Radial moments and density.**

Experiment 14.2 (Even moments)**.** For the 2,001,050 lag-two increments, Table 1 gives the empirical moments and the exact values from Proposition 3.2.

Table 1. Even radial moments of $|\Delta_{k,2}|$.

m	Empirical $\widehat{C}_m$	Exact C_m	Relative error
1	4.001000014	4	0.0002500
2	28.007876399	28	0.0002813
3	256.065581224	256	0.0002562
4	2716.554065884	2716	0.0002040
5	31509.108696389	31504	0.0001622

The empirical mean radius is

$$\frac{1}{2{,}001{,}050}\sum_k|\Delta_{k,2}| = 1.7994559491, \tag{14.1}$$

compared with the four-step reference value approximately 1.7990924798, consistent with the random-walk calculations in [2, 3].

14.2. **Phase means and the four correlations.**

Experiment 14.3 (Correlation decomposition)**.** The full-data means are shown in Table 2. The deterministic clock-clock term is smaller than the three terms containing true ordinates, as predicted by Theorem 8.3.

The mean residual phase itself is

$$\frac{1}{2{,}001{,}052}\sum_k b_k = -8.3396271\times10^{-6} - 1.0984572\times10^{-3}\mathrm{i}. \tag{14.2}$$

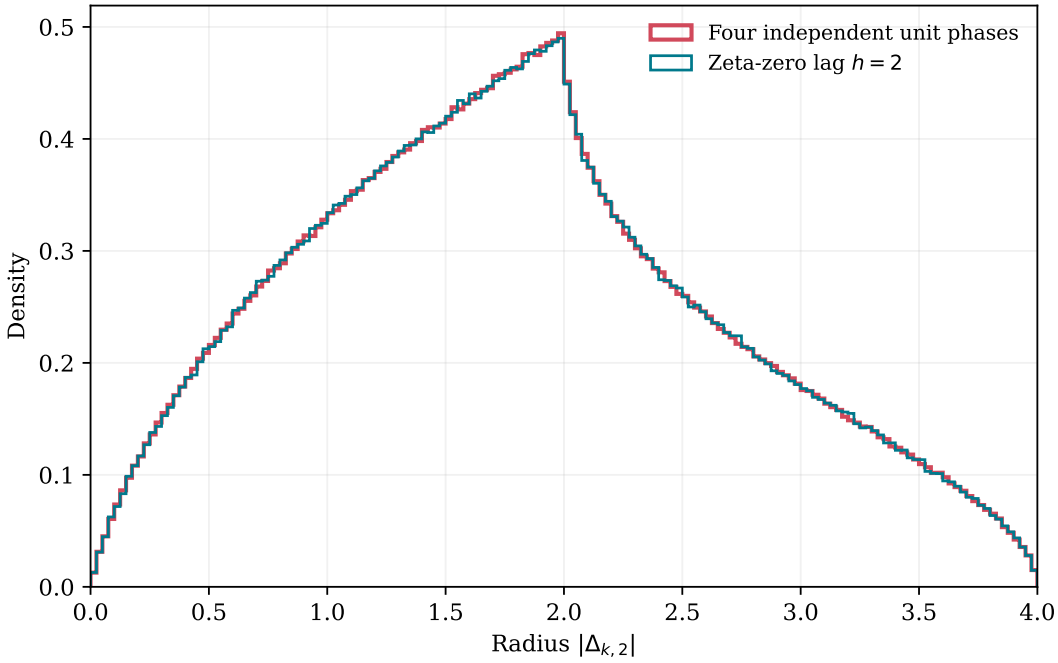


FIGURE 1. Empirical density of $|\Delta_{k,2}|$ for all lag-two increments, compared with a deterministic-seed Monte Carlo reference of four independent unit phases using 4,000,000 samples.

TABLE 2. Empirical lag-two correlations.

Correlation	Real part	Imaginary part
$\langle z_{k+2}\overline{z}_k\rangle$	$-5.1993826 \times 10^{-4}$	2.9360002×10^{-4}
$\langle z_{k+2}\overline{w}_k\rangle$	$-2.1722450 \times 10^{-4}$	5.0371940×10^{-4}
$\langle w_{k+2}\overline{z}_k\rangle$	1.6167899×10^{-4}	$-2.8085147 \times 10^{-4}$
$\langle w_{k+2}\overline{w}_k\rangle$	$-1.7059704 \times 10^{-5}$	1.7889030×10^{-5}

TABLE 3. RMS of block means for the two phase sequences.

L	RMS b_k	RMS $c_k^{(2)}$	$L^{-1/2}$
4	0.500125	0.499895	0.500000
8	0.353978	0.353624	0.353553
16	0.250311	0.250264	0.250000
32	0.176856	0.176473	0.176777
64	0.124813	0.124323	0.125000
128	0.087955	0.088057	0.088388
256	0.062228	0.062269	0.062500
512	0.043905	0.044154	0.044194
1024	0.030680	0.031107	0.031250
2048	0.021650	0.022053	0.022097
4096	0.015651	0.015720	0.015625

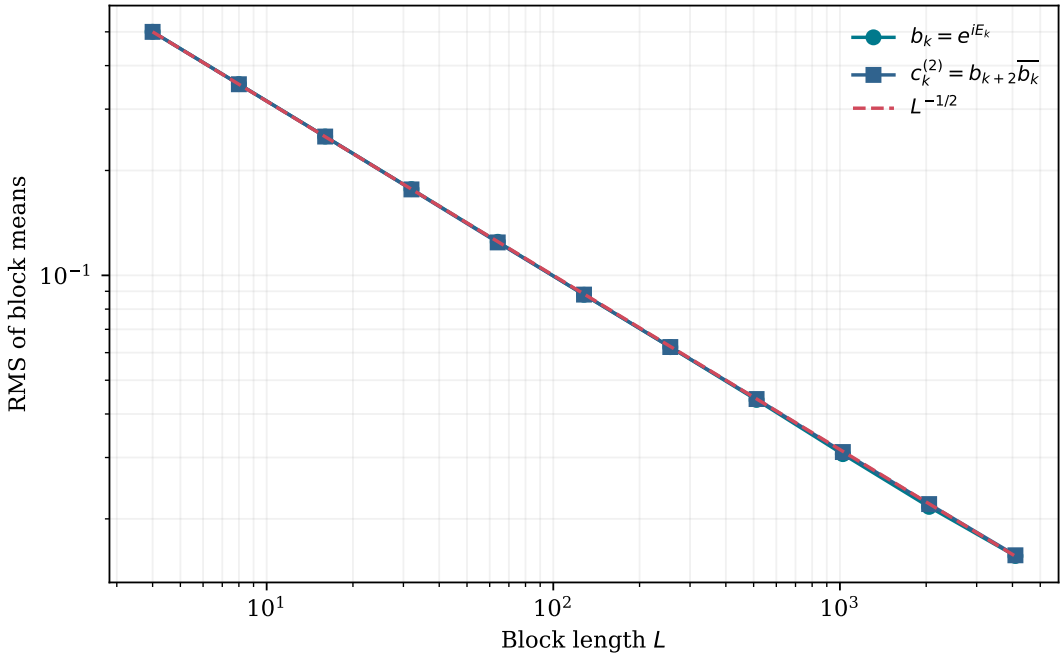


FIGURE 2. Block RMS for b_k, $c_k^{(2)}$, and the independent-phase benchmark $L^{-1/2}$.

14.3. **Block cancellation and angular uniformity.**

Experiment 14.4 (Block RMS)**.** Table 3 reports the root mean square of the complex block means for b_k and $c_k^{(2)}$. Both track the independent-phase benchmark $L^{-1/2}$ from $L = 4$ through $L = 4096$.

The empirical lag-two residual mean is

$$\frac{1}{2{,}001{,}050}\sum_k c_k^{(2)} = 3.3469204 \times 10^{-4} + 4.3859254 \times 10^{-4}\mathrm{i}, \tag{14.3}$$

whose modulus is 5.5170842×10^{-4}. The moduli of its first eight empirical angular Fourier coefficients are

$$\begin{aligned}(&0.000551708, 0.000130645, 0.000096481, 0.000191193,\\ &0.000576955, 0.000108608, 0.001008598, 0.000404103).\end{aligned} \tag{14.4}$$

For 32, 64, 128, and 256 equal angular sectors, the maximum occupancy divided by its uniform expectation is

$$(1.00579, 1.01345, 1.02992, 1.04009), \tag{14.5}$$

while the chi-square statistic divided by its degrees of freedom is

$$(0.6783, 0.8466, 1.1581, 1.0640). \tag{14.6}$$

The recurrence (12.2) was also evaluated directly; the maximum floating-point residual was 5.55×10^{-17}.

15. LITERATURE POSITION AND EXACT REMAINING THEOREM

The relevant literature separates into five layers.

(1) Riemann–von Mangoldt gives the smooth counter and its logarithmic remainder [18].

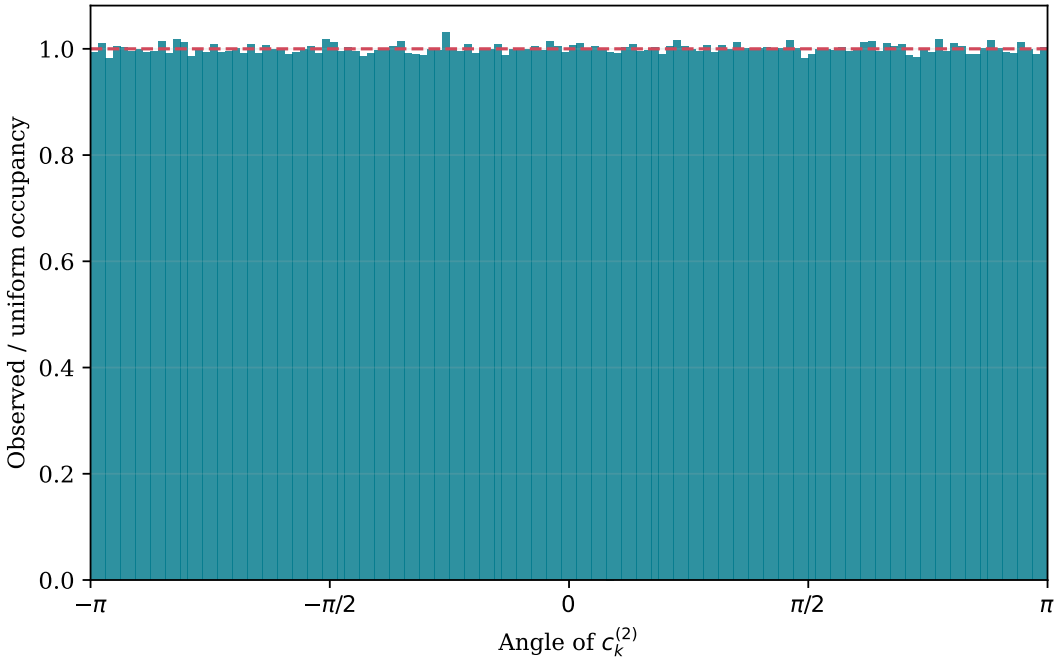


FIGURE 3. Angular occupancy of $c_k^{(2)}$ in 128 sectors, divided by the uniform expected occupancy.

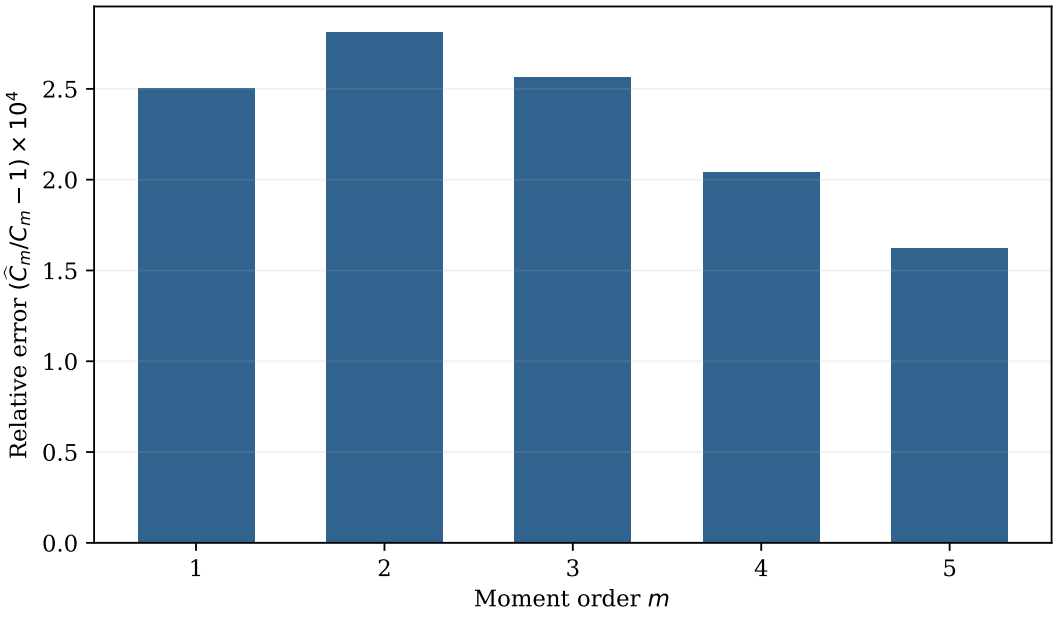


FIGURE 4. Relative errors of the first five even radial moments, in units of 10^{-4}.

(2) Hlawka–Fujii-type theorems give uniform distribution for fixed frequencies or fixed functions of the ordinates [11, 9, 5].
(3) Montgomery and Rudnick–Sarnak control pair and higher correlations in their test-function regimes [14, 17].

(4) Selberg-type and mesoscopic theorems describe coarse or normalized fluctuations [12, 4].
(5) Random-walk analysis supplies the exact target density and moments [2, 3, 13].

None of these layers supplies the triangular, fixed-neighbor estimate at frequency $q_k = k \log k$. The exact additional statement is not merely a pair-correlation or variance formula; it is the blockwise cancellation in (9.9) and (9.10), equivalently the superfine two-gap phase estimate (13.13) together with the single-phase analogue.

The scale-invisibility theorem explains why this gap is structural. A weak limit at resolution $1/\log k$ cannot see arbitrary phase changes generated at resolution $1/(k \log k)$. A successful proof must therefore use a source of arithmetic information finer than ordinary spacing convergence.

16. Conclusion

The lag-two shark-fin problem decomposes into a complete deterministic part and a sharply isolated arithmetic part. The deterministic clock correlation is proved to vanish by exact Lambert inversion and Kusmin–Landau cancellation. The second radial moment follows from two explicit mesoscopic phase estimates. The counter-band formula replaces critical-line contour displacement by a short consecutive index band. The apparent 4π resonance is broken by a calculable Lambert drift. Finally, the residual increment is expressed exactly using two consecutive normalized gaps and the counting remainder.

The numerical record from all 2,001,052 available ordinates is consistent across radial moments, block cancellation, angular Fourier modes, and histogram tests. The scale-invisibility theorem nevertheless shows why those observations and the standard weak gap laws do not automatically become an asymptotic proof. The remaining statement is a new superfine cancellation theorem for the actual zeros, at the natural phase resolution $1/(k \log k)$.

17. De quién viene esta idea

The four-step radial law, Weyl's criterion, the Riemann–von Mangoldt formula, Lambert W asymptotics, Kusmin–Landau estimates, pair correlation, and mesoscopic Gaussian fluctuation results are classical or literature-based and are cited at their points of use. The present construction selects the specific index frequency $q_k = k \log k$, subtracts the Lambert clock, and forms the lagged radius $|a_{k+2} - a_k|$.

The following parts are deductions specific to this analysis: the exact four-correlation decomposition for the chosen observable; the fixed-lag clock cancellation after removal of $2\pi k^2$; the mesoscopic transfer criterion; the oriented counter-band kernel; the identification of the internal 4π resonance and its Lambert detuning; the exact formula (12.5); and the scale-invisibility theorem adapted to $q_k = k \log k$. The empirical audit is specific to the user's complete Odlyzko-derived dataset and deterministic Lambert clock.

Appendix A. Reproducibility details

The binary files contain little-endian IEEE 754 double-precision values. The zero table and the clock table each contain 2,001,052 entries. For an array indexed from one, the computation is

$$q_k = k \log k, \tag{A.1}$$

$$E_k = q_k(\gamma_k - \tau_k), \tag{A.2}$$

$$a_k = \mathrm{e}^{\mathrm{i}q_k\gamma_k} - \mathrm{e}^{\mathrm{i}q_k\tau_k}, \tag{A.3}$$

$$\Delta_{k,2} = a_{k+2} - a_k, \tag{A.4}$$

$$c_k^{(2)} = \mathrm{e}^{\mathrm{i}(E_{k+2}-E_k)}. \tag{A.5}$$

Block RMS values use all complete nonoverlapping blocks and discard only the final incomplete block. Angular histograms use the principal argument in $[-\pi, \pi)$. The radial reference curve in

Figure 1 uses four million independent quadruples of uniform phases with a fixed pseudorandom seed.

References

[1] O. Bohigas, P. Leboeuf, and M. J. Sánchez, *Spectral spacing correlations for chaotic and disordered systems*, Found. Phys. **31** (2001), no. 3, 489–517. doi: 10.1023/A:1017569612944.

[2] J. M. Borwein, A. Straub, J. Wan, and W. Zudilin, *Densities of short uniform random walks*, Canad. J. Math. **64** (2012), no. 5, 961–990. doi: 10.4153/CJM-2011-079-2.

[3] J. M. Borwein, A. Straub, and J. Wan, *Three-step and four-step random walk integrals*, Experiment. Math. **22** (2013), no. 1, 1–14. doi: 10.1080/10586458.2013.748379.

[4] P. Bourgade, *Mesoscopic fluctuations of the zeta zeros*, Probab. Theory Related Fields **148** (2010), 479–500. doi: 10.1007/s00440-009-0237-3; arXiv:0902.1757.

[5] F. Çiçek and S. M. Gonek, *The uniform distribution modulo one of certain subsequences of ordinates of zeros of the zeta function*, Math. Proc. Cambridge Philos. Soc. **176** (2024), no. 3, 593–608. doi: 10.1017/S0305004124000045.

[6] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, *On the Lambert W function*, Adv. Comput. Math. **5** (1996), 329–359. doi: 10.1007/BF02124750.

[7] K. Ford, X. Meng, and A. Zaharescu, *Simultaneous distribution of the fractional parts of Riemann zeta zeros*, Bull. Lond. Math. Soc. **49** (2017), no. 1, 1–9. doi: 10.1112/blms.12007.

[8] K. Ford and A. Zaharescu, *On the distribution of imaginary parts of zeros of the Riemann zeta function*, J. Reine Angew. Math. **579** (2005), 145–158. doi: 10.1515/crll.2005.2005.579.145.

[9] A. Fujii, *On the uniformity of the distribution of the zeros of the Riemann zeta function*, J. Reine Angew. Math. **302** (1978), 167–205. doi: 10.1515/crll.1978.302.167.

[10] S. W. Graham and G. Kolesnik, *Van der Corput's Method of Exponential Sums*, London Math. Soc. Lecture Note Ser. **126**, Cambridge University Press, Cambridge, 1991. doi: 10.1017/CBO9780511661976.

[11] E. Hlawka, *Über die Gleichverteilung gewisser Folgen, welche mit den Nullstellen der Zetafunktion zusammenhängen*, Österreich. Akad. Wiss. Math.-Naturwiss. Kl. S.-B. II **184** (1975), no. 8–10, 459–471.

[12] V. Kargin, *On fluctuations of Riemann's zeta zeros*, Probab. Theory Related Fields **157** (2013), 575–604. doi: 10.1007/s00440-012-0465-9; arXiv:1203.5309.

[13] J. R. McCrorie, *Moments in Pearson's four-step uniform random walk problem and other applications of very well-poised generalized hypergeometric series*, Sankhyā B **83** (2021), Suppl. 2, 244–281. doi: 10.1007/s13571-020-00230-1.

[14] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in *Analytic Number Theory*, Proc. Sympos. Pure Math. **24**, Amer. Math. Soc., Providence, RI, 1973, 181–193.

[15] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Math. Comp. **48** (1987), no. 177, 273–308. doi: 10.1090/S0025-5718-1987-0866115-0.

[16] A. M. Odlyzko, *Tables of zeros of the Riemann zeta function*, University of Minnesota, public numerical tables, `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`, accessed 17 August 2026.

[17] Z. Rudnick and P. Sarnak, *Zeros of principal L-functions and random matrix theory*, Duke Math. J. **81** (1996), 269–322. doi: 10.1215/S0012-7094-96-08115-6.

[18] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, Oxford, 1986.

[19] H. Weyl, *Über die Gleichverteilung von Zahlen mod. Eins*, Math. Ann. **77** (1916), no. 3, 313–352. doi: 10.1007/BF01475864.

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